

Exercises Chapter 5

1. A researcher fits a logistic regression model to predict a binary outcome variable $Y \in \{0, 1\}$ using a single ($p = 1$) predictor variable X .

The values of the parameter estimates are $\hat{\beta}_0 = -2.4$ and $\hat{\beta}_1 = 0.8$.

- (a) Write down the fitted logistic regression model for $\hat{P}(Y = 1|X = x)$.
 - (b) Compute the value x of the predictor such that $\hat{P}(Y = 1|X = x) = \hat{P}(Y = 0|X = x)$.
2. Consider the following classifier f for a binary outcome variable $Y \in \{0, 1\}$ and a single ($p = 1$) predictor variable X ,

$$f(x_0) = \begin{cases} 1 & \text{if } P(Y = 1|X = x_0) \geq 0.5 \\ 0 & \text{if } P(Y = 1|X = x_0) < 0.5 \end{cases}$$

Let the true (usually unknown) conditional probability function be given by

$$P(Y = 1|X = x_0) = \begin{cases} 1 & \text{if } x_0 \geq 0.6 \\ 0 & \text{if } x_0 < 0.6 \end{cases}$$

You have a training data set of size $n_{\text{Train}} = 5$ generated according to the model above with $X_i = i/n_{\text{Train}}$:

$$\begin{aligned} & \{(X_1, Y_1), (X_2, Y_2), (X_3, Y_3), (X_4, Y_4), (X_5, Y_5)\} \\ &= \{(0.2, 0), (0.4, 0), (0.6, 1), (0.8, 1), (1, 1)\}. \end{aligned}$$

You estimate $P(Y = 1|X = x)$ using the KNN estimator

$$\hat{P}_K(Y = 1|X = x_0) = \frac{1}{K} \sum_{i \in \mathcal{N}_0} I(Y_i = 1),$$

applied to the training data set above. Given your estimation $\hat{P}_K(Y = 1|X = x_0)$ you construct the classifier $\hat{f}_K$ as above by replacing the true conditional probability by its estimate $\hat{P}_K(Y = 1|X = x_0)$.

- (a) What is the value of $\hat{P}_K(Y = 1|X = 0.4)$ for $K = 1$ and $K = 5$?
- (b) What is the training error rate for $K = 1$ and $K = 5$?